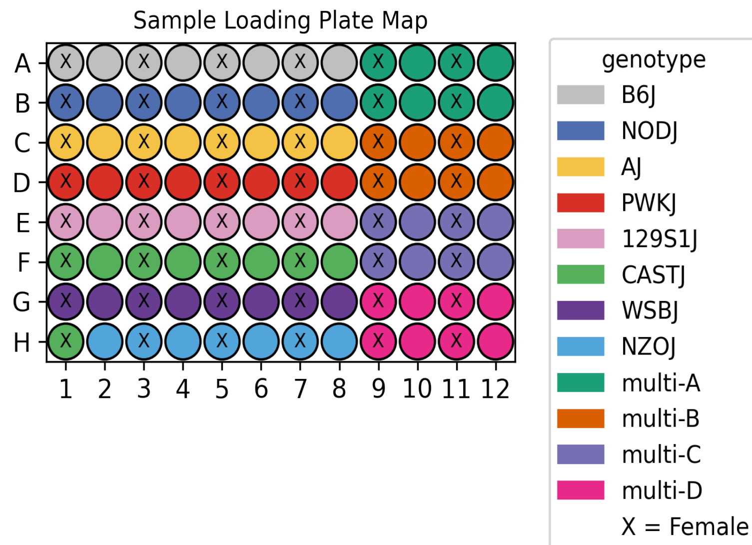
UCI/Caltech IGVF Dataset Pipeline Report

	igvf_007
Kit	WT_mega
Chemistry	Parse v2
Sample type	nuclei
Genotypes	B6J NODJ AJ PWKJ 129S1J CASTJ WSBJ NZOJ
Tissues	DiencephalonPituitary GonadsFemale GonadsMale
Subpool	Subpool_14
Measurment Set	IGVFDS5516XBYP
Exome capture subpool	No
Expected nuclei	67,000



1. Sample multiplexing

DiencephalonPituitary was the main tissue on the plate while GonadsMale and GonadsFemale were multiplexed.

Table 1: Genotype multiplexing key

Pair	Multiplexed Genotype 1	Multiplexed Genotype 2
multi-A	B6J	NODJ
multi-B	AJ	PWKJ
multi-C	129S1J	CASTJ
multi-D	NZOJ	WSBJ

2. Alignment

A total of 1,308,491,637 reads were aligned using GENCODE vM32 with mm39 assembly.

Table 2: Pseudoalignment stats

Subpool	Total reads	Total UMIs	Fraction duplicated UMIs	Total pseudoaligned	Pct. pseudoaligned
Subpool_14	1,308,491,637	685,570,853	0.241	903,528,292	69.1

3. Cell recovery

A total of 206,819 nuclei in Subpool_14 have >=200 UMIs and 82,185 nuclei have >= 500 UMIs.

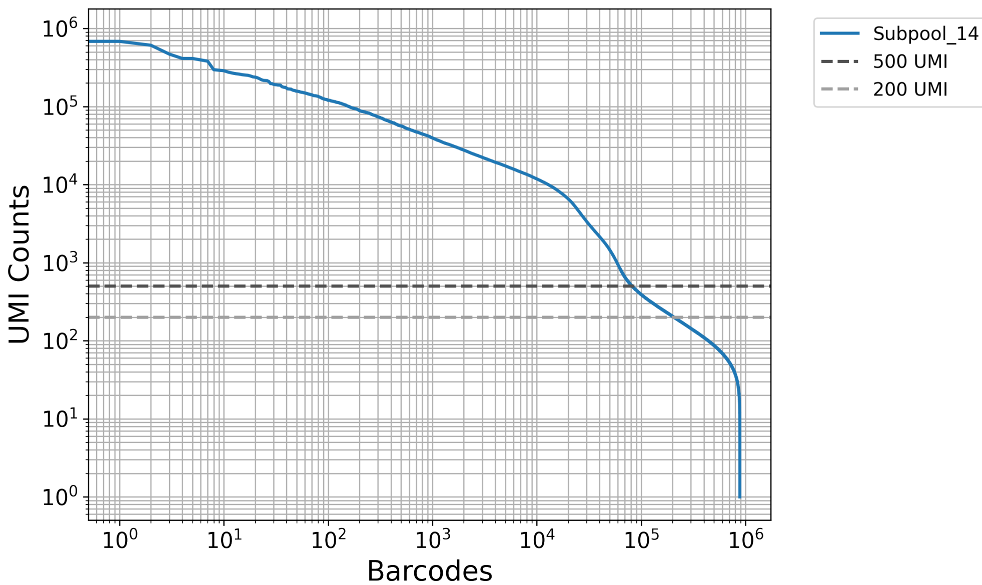


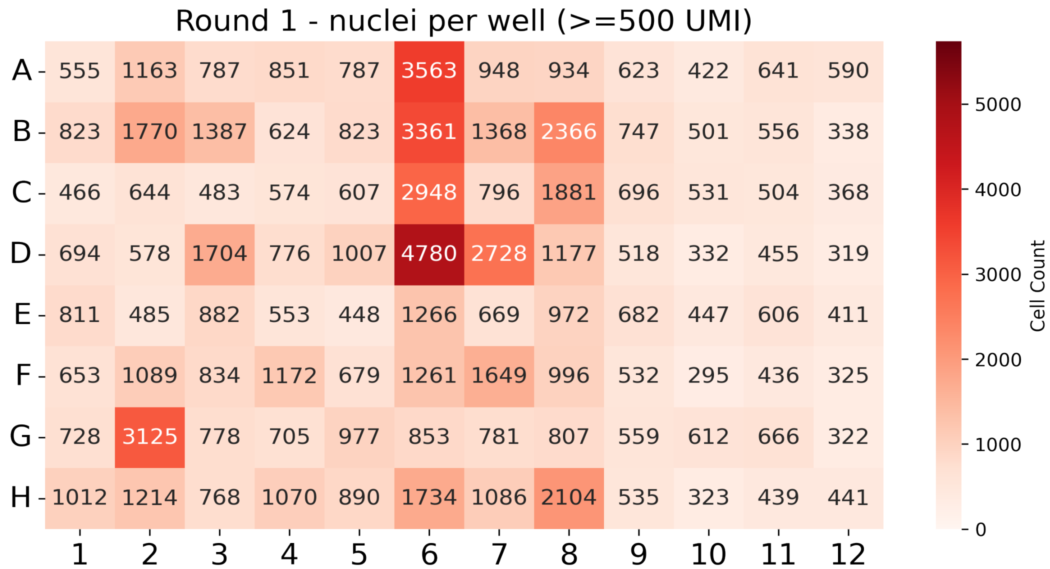
Table 3: QC stats for nuclei >=200 UMI

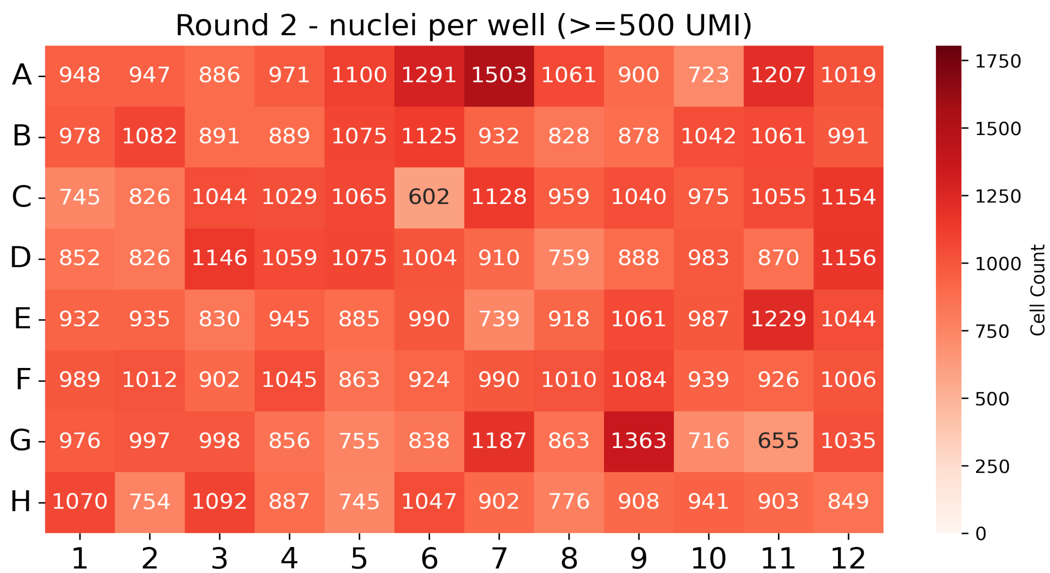
Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_14	206,819	377	2,354.5	318	910.2

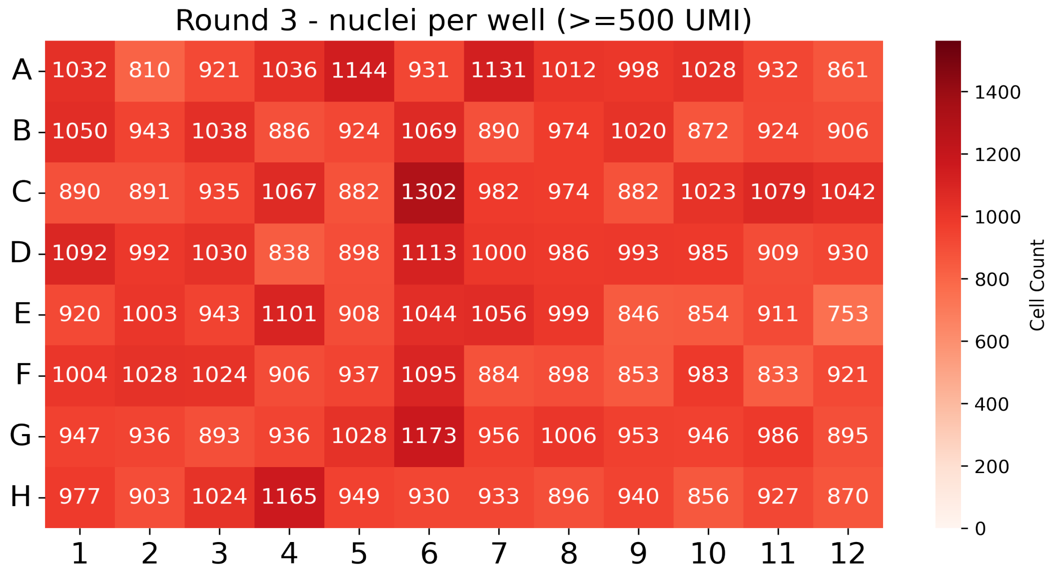
Table 4: QC stats for nuclei >=500 UMI

Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_14	82,185	2,063	5,478.4	1,268	1,906.3

In this combinatorial barcoding assay, nuclei are barcoded in 96-well plates over three rounds. The first round assigns sample-specific barcodes, while the next two are applied to the pooled mix. Even distribution in the first round is important to ensure consistent sample representation.







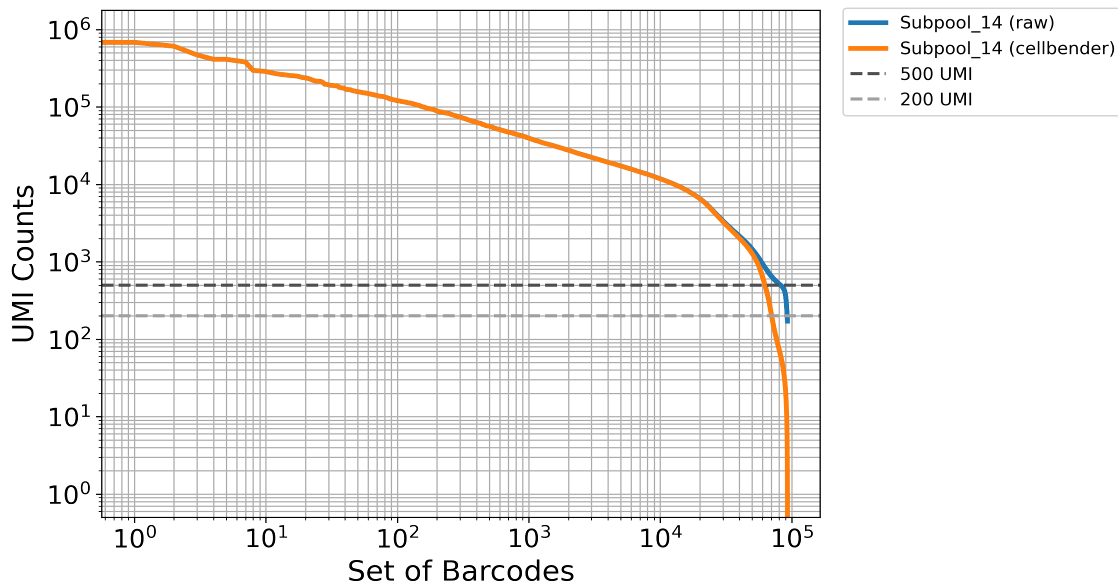
4. Background removal

Table 5: Cellbender parameters

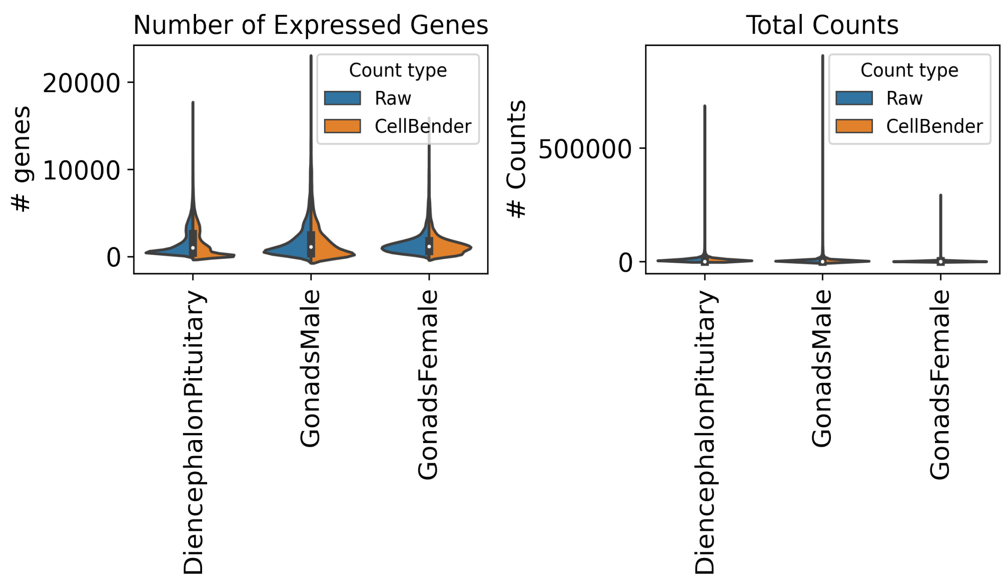
Subpool	droplets_included	learning_rate	expected_cells
Subpool_14	250000	5e-05	67000

Table 6: Cellbender stats

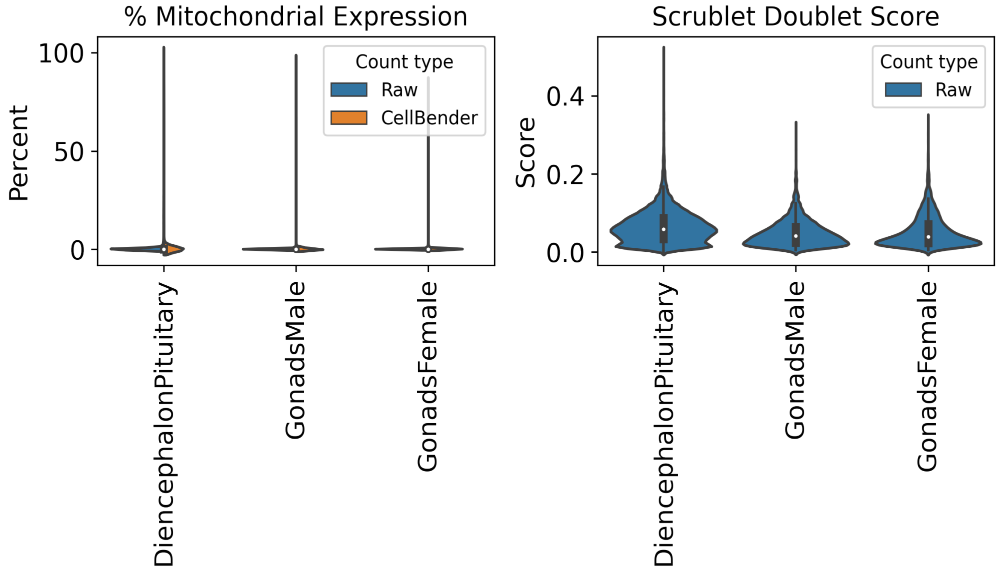
Subpool	Percent counts removed	Found nuclei	Mean denoised counts
Subpool_14	4.4	92,776	4,682.9



Subpool_14 (raw)
Subpool_14 (cellbender)
500 UMI
200 UMI
UMI Counts
Set of Barcodes



Number of Expressed Genes
genes
Count type
Raw
CellBender
DiencephalonPituitary
GonadsMale
GonadsFemale
Total Counts
Counts
Count type
Raw
CellBender
DiencephalonPituitary
GonadsMale
GonadsFemale



5. Barcode sample map

The first round corresponds to sample barcoding. In Parse Bio assays, two barcodes are included in the first round: a random hexamer (R) and oligo-dT (T) barcode per well (parse barcode type). This reduces 3-prime bias and allows for the capture of full-length transcripts. These sample level barcodes can be found in the barcode-1 region in the seqspec, and begin at position 16 (0-based) in the full 24-nt cell barcode. For the WT_mega kit, samples are distributed across 96 wells in round 1. The following table includes our lab sample ID as sample descriptions and IGVF sample accessions.

Table 7: Barcode sample map

barcode	sample accession	sample description	position	parse barcode type	well
CATCATCC	IGVFSM4465YOJN,IGVFSM5984CJUU	016_B6J_10F_01	16	R	A1
CATTCCTA	IGVFSM4465YOJN,IGVFSM5984CJUU	016_B6J_10F_01	16	T	A1
CTGCTTTG	IGVFSM5230YSVJ,IGVFSM6071JHCX	017_B6J_10M_01	16	R	A2
CTTCATCA	IGVFSM5230YSVJ,IGVFSM6071JHCX	017_B6J_10M_01	16	T	A2
CTAAGGGA	IGVFSM5188HZIG,IGVFSM9214NKTQ	018_B6J_10F_01	16	R	A3
CCTATATC	IGVFSM5188HZIG,IGVFSM9214NKTQ	018_B6J_10F_01	16	T	A3
GCTTATAG	IGVFSM0166IBCH,IGVFSM6699LJHC	019_B6J_10M_01	16	R	A4
ACATTTAC	IGVFSM0166IBCH,IGVFSM6699LJHC	019_B6J_10M_01	16	T	A4
TCTGATCC	IGVFSM3470VQGS,IGVFSM8528YUHU	020_B6J_10F_01	16	R	A5
ACTTAGCT	IGVFSM3470VQGS,IGVFSM8528YUHU	020_B6J_10F_01	16	T	A5

TCTCTTGG	IGVFSM7323JYDH,IGVFSM8952OBMF	021_B6J_10M_01	16	R	A6
CCAATTCT	IGVFSM7323JYDH,IGVFSM8952OBMF	021_B6J_10M_01	16	T	A6
CAATTTCC	IGVFSM1381VQIA,IGVFSM8656OHKM	024_B6J_10F_01	16	R	A7
GCCTATCT	IGVFSM1381VQIA,IGVFSM8656OHKM	024_B6J_10F_01	16	T	A7
AGTCTCTT	IGVFSM1156KMPC,IGVFSM4934KHPJ	025_B6J_10M_01	16	R	A8
ATGCTGCT	IGVFSM1156KMPC,IGVFSM4934KHPJ	025_B6J_10M_01	16	T	A8
TGCTGCTC	IGVFSM4567NUXK	016_B6J_10F_10	16	R	A9
CATTTACA	IGVFSM4567NUXK	016_B6J_10F_10	16	T	A9
TGCTGCTC	IGVFSM3537SUZG	016_B6J_10F_11	16	R	A9
CATTTACA	IGVFSM3537SUZG	016_B6J_10F_11	16	T	A9
TGCTGCTC	IGVFSM9963SJEX	066_NODJ_10F_10	16	R	A9
CATTTACA	IGVFSM9963SJEX	066_NODJ_10F_10	16	T	A9
TGCTGCTC	IGVFSM3417ZEGZ	066_NODJ_10F_11	16	R	A9
CATTTACA	IGVFSM3417ZEGZ	066_NODJ_10F_11	16	T	A9
GTATTTCC	IGVFSM6618MHBV	017_B6J_10M_10	16	R	A10
ACTCGTAA	IGVFSM6618MHBV	017_B6J_10M_10	16	T	A10
GTATTTCC	IGVFSM1349YMMU	017_B6J_10M_11	16	R	A10
ACTCGTAA	IGVFSM1349YMMU	017_B6J_10M_11	16	T	A10
GTATTTCC	IGVFSM8361NBRE	067_NODJ_10M_10	16	R	A10
ACTCGTAA	IGVFSM8361NBRE	067_NODJ_10M_10	16	T	A10
GTATTTCC	IGVFSM5322PFJO	067_NODJ_10M_11	16	R	A10
ACTCGTAA	IGVFSM5322PFJO	067_NODJ_10M_11	16	T	A10
TTCCTGTG	IGVFSM7587IQZO	018_B6J_10F_10	16	R	A11
CCTTTGCA	IGVFSM7587IQZO	018_B6J_10F_10	16	T	A11
TTCCTGTG	IGVFSM2910IGST	018_B6J_10F_11	16	R	A11
CCTTTGCA	IGVFSM2910IGST	018_B6J_10F_11	16	T	A11
TTCCTGTG	IGVFSM5649EBMW	068_NODJ_10F_10	16	R	A11
CCTTTGCA	IGVFSM5649EBMW	068_NODJ_10F_10	16	T	A11
TTCCTGTG	IGVFSM3199IWJM	068_NODJ_10F_11	16	R	A11
CCTTTGCA	IGVFSM3199IWJM	068_NODJ_10F_11	16	T	A11
GCTGCTTC	IGVFSM4552FSZN	019_B6J_10M_10	16	R	A12
ACTCCTGC	IGVFSM4552FSZN	019_B6J_10M_10	16	T	A12
GCTGCTTC	IGVFSM4635AIZP	019_B6J_10M_11	16	R	A12
ACTCCTGC	IGVFSM4635AIZP	019_B6J_10M_11	16	T	A12
GCTGCTTC	IGVFSM0379DQEB	069_NODJ_10M_10	16	R	A12

ACTCCTGC	IGVFSM0379DQEB	069_NODJ_10M_10	16	T	A12
GCTGCTTC	IGVFSM5988JHKG	069_NODJ_10M_11	16	R	A12
ACTCCTGC	IGVFSM5988JHKG	069_NODJ_10M_11	16	T	A12
TATGTGTC	IGVFSM3183PSGJ,IGVFSM6418OJGM	066_NODJ_10F_01	16	R	B1
ATTTGGCA	IGVFSM3183PSGJ,IGVFSM6418OJGM	066_NODJ_10F_01	16	T	B1
CAATTCTC	IGVFSM1910QUCB,IGVFSM6194RATY	067_NODJ_10M_01	16	R	B2
TTATTCTG	IGVFSM1910QUCB,IGVFSM6194RATY	067_NODJ_10M_01	16	T	B2
TGGTCTCC	IGVFSM4182XRQE,IGVFSM8455TNVO	068_NODJ_10F_01	16	R	B3
TCATGCTC	IGVFSM4182XRQE,IGVFSM8455TNVO	068_NODJ_10F_01	16	T	B3
GCTCTTTA	IGVFSM6842OMYX,IGVFSM8873UKSN	069_NODJ_10M_01	16	R	B4
CATACTTC	IGVFSM6842OMYX,IGVFSM8873UKSN	069_NODJ_10M_01	16	T	B4
GCTGCATG	IGVFSM0017ZTCP,IGVFSM1602MMIE	070_NODJ_10F_01	16	R	B5
CCGTTCTA	IGVFSM0017ZTCP,IGVFSM1602MMIE	070_NODJ_10F_01	16	T	B5
ACTCATTT	IGVFSM6636PLRM,IGVFSM8886HJDO	071_NODJ_10M_01	16	R	B6
GCTTCATA	IGVFSM6636PLRM,IGVFSM8886HJDO	071_NODJ_10M_01	16	T	B6
AGTCTTGG	IGVFSM0442MPEV,IGVFSM9173JXFM	072_NODJ_10F_01	16	R	B7
CTCTGTGC	IGVFSM0442MPEV,IGVFSM9173JXFM	072_NODJ_10F_01	16	T	B7
GGTTCTTC	IGVFSM0489RZXG,IGVFSM8174YDBO	073_NODJ_10M_01	16	R	B8
CCCTTATA	IGVFSM0489RZXG,IGVFSM8174YDBO	073_NODJ_10M_01	16	T	B8
TCATGTTG	IGVFSM6410RGFM	020_B6J_10F_10	16	R	B9
ACTGCTCT	IGVFSM6410RGFM	020_B6J_10F_10	16	T	B9
TCATGTTG	IGVFSM8699AZSF	020_B6J_10F_11	16	R	B9
ACTGCTCT	IGVFSM8699AZSF	020_B6J_10F_11	16	T	B9
TCATGTTG	IGVFSM6896ZGQR	070_NODJ_10F_10	16	R	B9
ACTGCTCT	IGVFSM6896ZGQR	070_NODJ_10F_10	16	T	B9
TCATGTTG	IGVFSM7472UIHU	070_NODJ_10F_11	16	R	B9
ACTGCTCT	IGVFSM7472UIHU	070_NODJ_10F_11	16	T	B9
ATTTTGCC	IGVFSM8837MYHY	021_B6J_10M_10	16	R	B10
CTCTAATC	IGVFSM8837MYHY	021_B6J_10M_10	16	T	B10
ATTTTGCC	IGVFSM5614XQUG	021_B6J_10M_11	16	R	B10
CTCTAATC	IGVFSM5614XQUG	021_B6J_10M_11	16	T	B10
ATTTTGCC	IGVFSM2896HLSZ	071_NODJ_10M_10	16	R	B10
CTCTAATC	IGVFSM2896HLSZ	071_NODJ_10M_10	16	T	B10
ATTTTGCC	IGVFSM3736YYMT	071_NODJ_10M_11	16	R	B10
CTCTAATC	IGVFSM3736YYMT	071_NODJ_10M_11	16	T	B10

CTTCTGTA	IGVFSM1095OHOH	024_B6J_10F_10	16	R	B11
ACCCTTGC	IGVFSM1095OHOH	024_B6J_10F_10	16	T	B11
CTTCTGTA	IGVFSM4730CIUG	024_B6J_10F_11	16	R	B11
ACCCTTGC	IGVFSM4730CIUG	024_B6J_10F_11	16	T	B11
CTTCTGTA	IGVFSM0625BIBT	072_NODJ_10F_10	16	R	B11
ACCCTTGC	IGVFSM0625BIBT	072_NODJ_10F_10	16	T	B11
CTTCTGTA	IGVFSM7300VPTD	072_NODJ_10F_11	16	R	B11
ACCCTTGC	IGVFSM7300VPTD	072_NODJ_10F_11	16	T	B11
GTCCATCT	IGVFSM3397IFNI	025_B6J_10M_10	16	R	B12
ATCTTAGG	IGVFSM3397IFNI	025_B6J_10M_10	16	T	B12
GTCCATCT	IGVFSM4481IPHX	025_B6J_10M_11	16	R	B12
ATCTTAGG	IGVFSM4481IPHX	025_B6J_10M_11	16	T	B12
GTCCATCT	IGVFSM4303UBDW	073_NODJ_10M_10	16	R	B12
ATCTTAGG	IGVFSM4303UBDW	073_NODJ_10M_10	16	T	B12
GTCCATCT	IGVFSM8768SXWV	073_NODJ_10M_11	16	R	B12
ATCTTAGG	IGVFSM8768SXWV	073_NODJ_10M_11	16	T	B12
GCTATCTC	IGVFSM2617YKKQ,IGVFSM8677EQEG	026_AJ_10F_01	16	R	C1
CATGTCTC	IGVFSM2617YKKQ,IGVFSM8677EQEG	026_AJ_10F_01	16	T	C1
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TCCATTAT	IGVFSM4796DKJO,IGVFSM4881USYE	028_AJ_10F_01	16	R	C3
ACACCTTT	IGVFSM4796DKJO,IGVFSM4881USYE	028_AJ_10F_01	16	T	C3
AGGATTAA	IGVFSM3351WODP,IGVFSM6893IKIH	031_AJ_10M_01	16	R	C4
AATTTCTC	IGVFSM3351WODP,IGVFSM6893IKIH	031_AJ_10M_01	16	T	C4
AATCTTTC	IGVFSM4955LORB,IGVFSM9545EZNW	030_AJ_10F_01	16	R	C5
ATTCATGG	IGVFSM4955LORB,IGVFSM9545EZNW	030_AJ_10F_01	16	T	C5
GTCATATG	IGVFSM1755SWJI,IGVFSM2119ANJP	033_AJ_10M_01	16	R	C6
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CTTCTAAC	IGVFSM3957TLDF,IGVFSM6828FNBA	032_AJ_10F_01	16	T	C7
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CCATCTTG	IGVFSM7508KQHS	026_AJ_10F_10	16	R	C9
TCTCATGC	IGVFSM7508KQHS	026_AJ_10F_10	16	T	C9
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TACTGTCT	IGVFSM9563RZBJ	029_AJ_10M_11	16	R	C10
ATCCTTAC	IGVFSM9563RZBJ	029_AJ_10M_11	16	T	C10
TACTGTCT	IGVFSM4307KCOE	077_PWKJ_10M_10	16	R	C10
ATCCTTAC	IGVFSM4307KCOE	077_PWKJ_10M_10	16	T	C10
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ATCCTTAC	IGVFSM5586TLOK	077_PWKJ_10M_11	16	T	C10
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TTCATCGC	IGVFSM9366IMCS	028_AJ_10F_11	16	R	C11
TAAATATC	IGVFSM9366IMCS	028_AJ_10F_11	16	T	C11
TTCATCGC	IGVFSM5409UMED	078_PWKJ_10F_10	16	R	C11
TAAATATC	IGVFSM5409UMED	078_PWKJ_10F_10	16	T	C11
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ACTGTGGG	IGVFSM5117TLYJ	031_AJ_10M_11	16	R	C12
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ACTGTGGG	IGVFSM3239STTL	079_PWKJ_10M_10	16	R	C12
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ACTGTGGG	IGVFSM2227UKRP	079_PWKJ_10M_11	16	R	C12
TTACCTGC	IGVFSM2227UKRP	079_PWKJ_10M_11	16	T	C12
TCTGTGCC	IGVFSM5417LNGC,IGVFSM6924RRKD	076_PWKJ_10F_01	16	R	D1
CACTTTCA	IGVFSM5417LNGC,IGVFSM6924RRKD	076_PWKJ_10F_01	16	T	D1
TCAATCTC	IGVFSM4103LTZB,IGVFSM5500OUCV	077_PWKJ_10M_01	16	R	D2
CACCTTTA	IGVFSM4103LTZB,IGVFSM5500OUCV	077_PWKJ_10M_01	16	T	D2
GTCCTCTG	IGVFSM2146SQAW,IGVFSM9111MIVH	078_PWKJ_10F_01	16	R	D3
CTGACTTC	IGVFSM2146SQAW,IGVFSM9111MIVH	078_PWKJ_10F_01	16	T	D3

TTACATTC	IGVFSM3536WDNR,IGVFSM3744QPBB	079_PWKJ_10M_01	16	R	D4
CATTTGGA	IGVFSM3536WDNR,IGVFSM3744QPBB	079_PWKJ_10M_01	16	T	D4
ATTCTGTC	IGVFSM0567BKYP,IGVFSM1552JBSI	080_PWKJ_10F_01	16	R	D5
GCTCTACT	IGVFSM0567BKYP,IGVFSM1552JBSI	080_PWKJ_10F_01	16	T	D5
TGTGTATG	IGVFSM6086KVUS,IGVFSM8490BJOU	081_PWKJ_10M_01	16	R	D6
GTTACGTA	IGVFSM6086KVUS,IGVFSM8490BJOU	081_PWKJ_10M_01	16	T	D6
TCCATTTG	IGVFSM4249JBEE,IGVFSM7955ABKB	084_PWKJ_10F_01	16	R	D7
CCTGTTGC	IGVFSM4249JBEE,IGVFSM7955ABKB	084_PWKJ_10F_01	16	T	D7
TTAGCTTC	IGVFSM3050OPDO,IGVFSM4510XDYL	085_PWKJ_10M_01	16	R	D8
CTATCATC	IGVFSM3050OPDO,IGVFSM4510XDYL	085_PWKJ_10M_01	16	T	D8
GTGCTTGA	IGVFSM6905VGKP	030_AJ_10F_10	16	R	D9
GCTATCAT	IGVFSM6905VGKP	030_AJ_10F_10	16	T	D9
GTGCTTGA	IGVFSM1408BMPX	030_AJ_10F_11	16	R	D9
GCTATCAT	IGVFSM1408BMPX	030_AJ_10F_11	16	T	D9
GTGCTTGA	IGVFSM2642TMHC	080_PWKJ_10F_10	16	R	D9
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GTGCTTGA	IGVFSM6638YNDH	080_PWKJ_10F_11	16	R	D9
GCTATCAT	IGVFSM6638YNDH	080_PWKJ_10F_11	16	T	D9
GTTTGTGA	IGVFSM1633EQRW	033_AJ_10M_10	16	R	D10
ACATTCAT	IGVFSM1633EQRW	033_AJ_10M_10	16	T	D10
GTTTGTGA	IGVFSM0075HJSS	033_AJ_10M_11	16	R	D10
ACATTCAT	IGVFSM0075HJSS	033_AJ_10M_11	16	T	D10
GTTTGTGA	IGVFSM3142HYPK	081_PWKJ_10M_10	16	R	D10
ACATTCAT	IGVFSM3142HYPK	081_PWKJ_10M_10	16	T	D10
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ACATTCAT	IGVFSM1277GJRQ	081_PWKJ_10M_11	16	T	D10
GAAATTAG	IGVFSM1911LHOP	032_AJ_10F_10	16	R	D11
TTCGCTAC	IGVFSM1911LHOP	032_AJ_10F_10	16	T	D11
GAAATTAG	IGVFSM2604ZWZC	032_AJ_10F_11	16	R	D11
TTCGCTAC	IGVFSM2604ZWZC	032_AJ_10F_11	16	T	D11
GAAATTAG	IGVFSM6801NQEI	084_PWKJ_10F_10	16	R	D11
TTCGCTAC	IGVFSM6801NQEI	084_PWKJ_10F_10	16	T	D11
GAAATTAG	IGVFSM9469AVQG	084_PWKJ_10F_11	16	R	D11
TTCGCTAC	IGVFSM9469AVQG	084_PWKJ_10F_11	16	T	D11
GCAAATTC	IGVFSM2230WLPO	035_AJ_10M_10	16	R	D12

CATTCTAC	IGVFSM2230WLPO	035_AJ_10M_10	16	T	D12
GCAAATTC	IGVFSM7820FOJS	035_AJ_10M_11	16	R	D12
CATTCTAC	IGVFSM7820FOJS	035_AJ_10M_11	16	T	D12
GCAAATTC	IGVFSM5592JXHF	085_PWKJ_10M_10	16	R	D12
CATTCTAC	IGVFSM5592JXHF	085_PWKJ_10M_10	16	T	D12
GCAAATTC	IGVFSM1980ATRP	085_PWKJ_10M_11	16	R	D12
CATTCTAC	IGVFSM1980ATRP	085_PWKJ_10M_11	16	T	D12
GAGGTTGA	IGVFSM4167SEXH,IGVFSM9398YHXY	036_129S1J_10F_01	16	R	E1
CACTTATC	IGVFSM4167SEXH,IGVFSM9398YHXY	036_129S1J_10F_01	16	T	E1
CCTGTCTG	IGVFSM3753OVHM,IGVFSM9806TYXF	037_129S1J_10M_01	16	R	E2
ATAAGCTC	IGVFSM3753OVHM,IGVFSM9806TYXF	037_129S1J_10M_01	16	T	E2
GTGGGTTC	IGVFSM5458IIQG,IGVFSM9530NJWB	038_129S1J_10F_01	16	R	E3
TCATCCTG	IGVFSM5458IIQG,IGVFSM9530NJWB	038_129S1J_10F_01	16	T	E3
TTTGCATC	IGVFSM0340DRJV,IGVFSM4269MUAL	041_129S1J_10M_01	16	R	E4
CCTGGTAT	IGVFSM0340DRJV,IGVFSM4269MUAL	041_129S1J_10M_01	16	T	E4
AGGTAATA	IGVFSM6249RJQV,IGVFSM7479OZRI	040_129S1J_10F_01	16	R	E5
TGGTATAC	IGVFSM6249RJQV,IGVFSM7479OZRI	040_129S1J_10F_01	16	T	E5
GTGCCTTC	IGVFSM6782GARZ,IGVFSM8930SAOK	043_129S1J_10M_01	16	R	E6
TTGGGAGA	IGVFSM6782GARZ,IGVFSM8930SAOK	043_129S1J_10M_01	16	T	E6
ATGTTTCC	IGVFSM0539MTAN,IGVFSM6738OHZO	044_129S1J_10F_01	16	R	E7
ACTTCATC	IGVFSM0539MTAN,IGVFSM6738OHZO	044_129S1J_10F_01	16	T	E7
CTTAATTC	IGVFSM1996CIIR,IGVFSM6742HLKN	045_129S1J_10M_01	16	R	E8
TCTCTAGC	IGVFSM1996CIIR,IGVFSM6742HLKN	045_129S1J_10M_01	16	T	E8
TCTGGCTC	IGVFSM6816QWSB	036_129S1J_10F_10	16	R	E9
ATGCCCTT	IGVFSM6816QWSB	036_129S1J_10F_10	16	T	E9
TCTGGCTC	IGVFSM8822PZTL	036_129S1J_10F_11	16	R	E9
ATGCCCTT	IGVFSM8822PZTL	036_129S1J_10F_11	16	T	E9
TCTGGCTC	IGVFSM7969ZAUZ	086_CASTJ_10F_10	16	R	E9
ATGCCCTT	IGVFSM7969ZAUZ	086_CASTJ_10F_10	16	T	E9
TCTGGCTC	IGVFSM8330OVCV	086_CASTJ_10F_11	16	R	E9
ATGCCCTT	IGVFSM8330OVCV	086_CASTJ_10F_11	16	T	E9
CATCATTT	IGVFSM1094VAAI	037_129S1J_10M_10	16	R	E10
CCCAATTT	IGVFSM1094VAAI	037_129S1J_10M_10	16	T	E10
CATCATTT	IGVFSM9038UIJU	037_129S1J_10M_11	16	R	E10
CCCAATTT	IGVFSM9038UIJU	037_129S1J_10M_11	16	T	E10

CATCATTT	IGVFSM3300FHLI	087_CASTJ_10M_10	16	R	E10
CCCAATTT	IGVFSM3300FHLI	087_CASTJ_10M_10	16	T	E10
CATCATTT	IGVFSM4100ZICM	087_CASTJ_10M_11	16	R	E10
CCCAATTT	IGVFSM4100ZICM	087_CASTJ_10M_11	16	T	E10
GTTGTCTC	IGVFSM2570ENQI	038_129S1J_10F_10	16	R	E11
ACTATATA	IGVFSM2570ENQI	038_129S1J_10F_10	16	T	E11
GTTGTCTC	IGVFSM1263AYLV	038_129S1J_10F_11	16	R	E11
ACTATATA	IGVFSM1263AYLV	038_129S1J_10F_11	16	T	E11
GTTGTCTC	IGVFSM0900XHGX	088_CASTJ_10F_10	16	R	E11
ACTATATA	IGVFSM0900XHGX	088_CASTJ_10F_10	16	T	E11
GTTGTCTC	IGVFSM2098KNMX	088_CASTJ_10F_11	16	R	E11
ACTATATA	IGVFSM2098KNMX	088_CASTJ_10F_11	16	T	E11
ATCTTCTG	IGVFSM8496DJXG	041_129S1J_10M_10	16	R	E12
CTCTATAC	IGVFSM8496DJXG	041_129S1J_10M_10	16	T	E12
ATCTTCTG	IGVFSM9236NDAG	041_129S1J_10M_11	16	R	E12
CTCTATAC	IGVFSM9236NDAG	041_129S1J_10M_11	16	T	E12
ATCTTCTG	IGVFSM7381JEDM	089_CASTJ_10M_10	16	R	E12
CTCTATAC	IGVFSM7381JEDM	089_CASTJ_10M_10	16	T	E12
ATCTTCTG	IGVFSM9723WBIN	089_CASTJ_10M_11	16	R	E12
CTCTATAC	IGVFSM9723WBIN	089_CASTJ_10M_11	16	T	E12
TGTTTGCC	IGVFSM7092FDTP,IGVFSM9286SBAR	086_CASTJ_10F_01	16	R	F1
CTGTCTCA	IGVFSM7092FDTP,IGVFSM9286SBAR	086_CASTJ_10F_01	16	T	F1
TTCTGTCA	IGVFSM4956HIAH,IGVFSM5457PTKY	087_CASTJ_10M_01	16	R	F2
GACCTTTC	IGVFSM4956HIAH,IGVFSM5457PTKY	087_CASTJ_10M_01	16	T	F2
ACGGACTC	IGVFSM0690VJRZ,IGVFSM9111THIZ	088_CASTJ_10F_01	16	R	F3
GATTTGGC	IGVFSM0690VJRZ,IGVFSM9111THIZ	088_CASTJ_10F_01	16	T	F3
TTTGGTCA	IGVFSM6740OVAR,IGVFSM8458TCXX	089_CASTJ_10M_01	16	R	F4
CGTCTAGG	IGVFSM6740OVAR,IGVFSM8458TCXX	089_CASTJ_10M_01	16	T	F4
TATCCGGG	IGVFSM1264GFRJ,IGVFSM5214BQRA	090_CASTJ_10F_01	16	R	F5
TACTCGAA	IGVFSM1264GFRJ,IGVFSM5214BQRA	090_CASTJ_10F_01	16	T	F5
TGTCATTC	IGVFSM4321DFFH,IGVFSM8357WYSN	091_CASTJ_10M_01	16	R	F6
CAGCCTTT	IGVFSM4321DFFH,IGVFSM8357WYSN	091_CASTJ_10M_01	16	T	F6
ATTCTCTG	IGVFSM0021WQYA,IGVFSM1257AOMZ	094_CASTJ_10F_01	16	R	F7
CCTCATTA	IGVFSM0021WQYA,IGVFSM1257AOMZ	094_CASTJ_10F_01	16	T	F7
TGGCTTCC	IGVFSM1436ECIH,IGVFSM5017BJEE	093_CASTJ_10M_01	16	R	F8

CTTATACC	IGVFSM1436ECIH,IGVFSM5017BJEE	093_CASTJ_10M_01	16	T	F8
TTGTTGCC	IGVFSM7330MCOB	040_129S1J_10F_10	16	R	F9
TCTATTAC	IGVFSM7330MCOB	040_129S1J_10F_10	16	T	F9
TTGTTGCC	IGVFSM9813UIKP	040_129S1J_10F_11	16	R	F9
TCTATTAC	IGVFSM9813UIKP	040_129S1J_10F_11	16	T	F9
TTGTTGCC	IGVFSM5920YOJY	090_CASTJ_10F_10	16	R	F9
TCTATTAC	IGVFSM5920YOJY	090_CASTJ_10F_10	16	T	F9
TTGTTGCC	IGVFSM1103DUUT	090_CASTJ_10F_11	16	R	F9
TCTATTAC	IGVFSM1103DUUT	090_CASTJ_10F_11	16	T	F9
GTCATCTC	IGVFSM6313PLBW	043_129S1J_10M_10	16	R	F10
CCTGCATT	IGVFSM6313PLBW	043_129S1J_10M_10	16	T	F10
GTCATCTC	IGVFSM9650VFSK	043_129S1J_10M_11	16	R	F10
CCTGCATT	IGVFSM9650VFSK	043_129S1J_10M_11	16	T	F10
GTCATCTC	IGVFSM0628FIYW	091_CASTJ_10M_10	16	R	F10
CCTGCATT	IGVFSM0628FIYW	091_CASTJ_10M_10	16	T	F10
GTCATCTC	IGVFSM6763UWGN	091_CASTJ_10M_11	16	R	F10
CCTGCATT	IGVFSM6763UWGN	091_CASTJ_10M_11	16	T	F10
TTGCTCAT	IGVFSM4841JIOO	044_129S1J_10F_10	16	R	F11
CAATCCTT	IGVFSM4841JIOO	044_129S1J_10F_10	16	T	F11
TTGCTCAT	IGVFSM5851CIQU	044_129S1J_10F_11	16	R	F11
CAATCCTT	IGVFSM5851CIQU	044_129S1J_10F_11	16	T	F11
TTGCTCAT	IGVFSM6711SYYK	094_CASTJ_10F_10	16	R	F11
CAATCCTT	IGVFSM6711SYYK	094_CASTJ_10F_10	16	T	F11
TTGCTCAT	IGVFSM8588WXUX	094_CASTJ_10F_11	16	R	F11
CAATCCTT	IGVFSM8588WXUX	094_CASTJ_10F_11	16	T	F11
CTGTCTGC	IGVFSM7262NIVF	045_129S1J_10M_10	16	R	F12
TTGTCTTA	IGVFSM7262NIVF	045_129S1J_10M_10	16	T	F12
CTGTCTGC	IGVFSM0998LUIZ	045_129S1J_10M_11	16	R	F12
TTGTCTTA	IGVFSM0998LUIZ	045_129S1J_10M_11	16	T	F12
CTGTCTGC	IGVFSM9677NTDJ	093_CASTJ_10M_10	16	R	F12
TTGTCTTA	IGVFSM9677NTDJ	093_CASTJ_10M_10	16	T	F12
CTGTCTGC	IGVFSM7066GBZS	093_CASTJ_10M_11	16	R	F12
TTGTCTTA	IGVFSM7066GBZS	093_CASTJ_10M_11	16	T	F12
TATATTCC	IGVFSM0966WWIZ,IGVFSM4130KIEV	058_WSBJ_10F_01	16	R	G1
TCACTTTA	IGVFSM0966WWIZ,IGVFSM4130KIEV	058_WSBJ_10F_01	16	T	G1

ATATTGGC	IGVFSM7450WJIV,IGVFSM8498UCVA	057_WSBJ_10M_01	16	R	G2
TGCTTGGG	IGVFSM7450WJIV,IGVFSM8498UCVA	057_WSBJ_10M_01	16	T	G2
GTGTCCTC	IGVFSM0752ZRMG,IGVFSM3371TFMC	060_WSBJ_10F_01	16	R	G3
CGCTCATT	IGVFSM0752ZRMG,IGVFSM3371TFMC	060_WSBJ_10F_01	16	T	G3
ATCTTCAT	IGVFSM2287EHGA,IGVFSM9356QKBM	059_WSBJ_10M_01	16	R	G4
GCCTCTAT	IGVFSM2287EHGA,IGVFSM9356QKBM	059_WSBJ_10M_01	16	T	G4
CGTGGTTG	IGVFSM2047VYYJ,IGVFSM7560RVGB	062_WSBJ_10F_01	16	R	G5
GAGCACAA	IGVFSM2047VYYJ,IGVFSM7560RVGB	062_WSBJ_10F_01	16	T	G5
TTGCATCC	IGVFSM3364FXFB,IGVFSM4632PCNC	061_WSBJ_10M_01	16	R	G6
CTCTTAAC	IGVFSM3364FXFB,IGVFSM4632PCNC	061_WSBJ_10M_01	16	T	G6
TCTTAATC	IGVFSM1348QFIK,IGVFSM6620ZEBK	096_WSBJ_10F_01	16	R	G7
TCTAGGCT	IGVFSM1348QFIK,IGVFSM6620ZEBK	096_WSBJ_10F_01	16	T	G7
TGCATTTC	IGVFSM7877XJMN,IGVFSM9074ZTED	063_WSBJ_10M_01	16	R	G8
AATTCTGC	IGVFSM7877XJMN,IGVFSM9074ZTED	063_WSBJ_10M_01	16	T	G8
GATGTTTC	IGVFSM6124VCIY	046_NZOJ_10F_10	16	R	G9
CATTCTCA	IGVFSM6124VCIY	046_NZOJ_10F_10	16	T	G9
GATGTTTC	IGVFSM8844CIBF	046_NZOJ_10F_11	16	R	G9
CATTCTCA	IGVFSM8844CIBF	046_NZOJ_10F_11	16	T	G9
GATGTTTC	IGVFSM8653IGLV	058_WSBJ_10F_10	16	R	G9
CATTCTCA	IGVFSM8653IGLV	058_WSBJ_10F_10	16	T	G9
GATGTTTC	IGVFSM7260RGBF	058_WSBJ_10F_11	16	R	G9
CATTCTCA	IGVFSM7260RGBF	058_WSBJ_10F_11	16	T	G9
ATCTTGTC	IGVFSM5941JXID	047_NZOJ_10M_10	16	R	G10
ACTTGCCT	IGVFSM5941JXID	047_NZOJ_10M_10	16	T	G10
ATCTTGTC	IGVFSM2925KIGE	047_NZOJ_10M_11	16	R	G10
ACTTGCCT	IGVFSM2925KIGE	047_NZOJ_10M_11	16	T	G10
ATCTTGTC	IGVFSM6049JASP	057_WSBJ_10M_10	16	R	G10
ACTTGCCT	IGVFSM6049JASP	057_WSBJ_10M_10	16	T	G10
ATCTTGTC	IGVFSM8555WNHX	057_WSBJ_10M_11	16	R	G10
ACTTGCCT	IGVFSM8555WNHX	057_WSBJ_10M_11	16	T	G10
TCATATTC	IGVFSM3735WYFG	048_NZOJ_10F_10	16	R	G11
ATCATTGC	IGVFSM3735WYFG	048_NZOJ_10F_10	16	T	G11
TCATATTC	IGVFSM7325IMAE	048_NZOJ_10F_11	16	R	G11
ATCATTGC	IGVFSM7325IMAE	048_NZOJ_10F_11	16	T	G11
TCATATTC	IGVFSM1827LDDT	060_WSBJ_10F_10	16	R	G11

ATCATTGC	IGVFSM1827LDDT	060_WSBJ_10F_10	16	T	G11
TCATATTC	IGVFSM5070ERGT	060_WSBJ_10F_11	16	R	G11
ATCATTGC	IGVFSM5070ERGT	060_WSBJ_10F_11	16	T	G11
TGGCCTCT	IGVFSM5754TPRZ	049_NZOJ_10M_10	16	R	G12
GTTCAACA	IGVFSM5754TPRZ	049_NZOJ_10M_10	16	T	G12
TGGCCTCT	IGVFSM6425UPLJ	049_NZOJ_10M_11	16	R	G12
GTTCAACA	IGVFSM6425UPLJ	049_NZOJ_10M_11	16	T	G12
TGGCCTCT	IGVFSM0997ZFEX	059_WSBJ_10M_10	16	R	G12
GTTCAACA	IGVFSM0997ZFEX	059_WSBJ_10M_10	16	T	G12
TGGCCTCT	IGVFSM6693YSWV	059_WSBJ_10M_11	16	R	G12
GTTCAACA	IGVFSM6693YSWV	059_WSBJ_10M_11	16	T	G12
CGTTGTCT	IGVFSM1078KTYU,IGVFSM8134HPUH	092_CASTJ_10F_01	16	R	H1
CCATTTGC	IGVFSM1078KTYU,IGVFSM8134HPUH	092_CASTJ_10F_01	16	T	H1
TCTTGTCA	IGVFSM4107ZNSE,IGVFSM6703AANL	047_NZOJ_10M_01	16	R	H2
GACTTTGC	IGVFSM4107ZNSE,IGVFSM6703AANL	047_NZOJ_10M_01	16	T	H2
TATTCCTG	IGVFSM0298LFOQ,IGVFSM8754ELWG	048_NZOJ_10F_01	16	R	H3
ATTGGCTC	IGVFSM0298LFOQ,IGVFSM8754ELWG	048_NZOJ_10F_01	16	T	H3
TCCATGTC	IGVFSM2572HXXE,IGVFSM9197PRMM	049_NZOJ_10M_01	16	R	H4
GTGCTAGC	IGVFSM2572HXXE,IGVFSM9197PRMM	049_NZOJ_10M_01	16	T	H4
TTGTCATC	IGVFSM2120LYVH,IGVFSM8205AVFV	050_NZOJ_10F_01	16	R	H5
CTTTCAAC	IGVFSM2120LYVH,IGVFSM8205AVFV	050_NZOJ_10F_01	16	T	H5
ATTTCCTG	IGVFSM3556PAEN,IGVFSM9711JMMI	051_NZOJ_10M_01	16	R	H6
ACTATTGC	IGVFSM3556PAEN,IGVFSM9711JMMI	051_NZOJ_10M_01	16	T	H6
GTGTCTCC	IGVFSM5374RXFR,IGVFSM9813BNZU	052_NZOJ_10F_01	16	R	H7
ACTGGCTT	IGVFSM5374RXFR,IGVFSM9813BNZU	052_NZOJ_10F_01	16	T	H7
GTGTGTGT	IGVFSM3167TBKE,IGVFSM9817PYLC	053_NZOJ_10M_01	16	R	H8
ATTAGGCT	IGVFSM3167TBKE,IGVFSM9817PYLC	053_NZOJ_10M_01	16	T	H8
TATGCTTC	IGVFSM1713ZBFY	050_NZOJ_10F_10	16	R	H9
GCCTTTCA	IGVFSM1713ZBFY	050_NZOJ_10F_10	16	T	H9
TATGCTTC	IGVFSM4889EESW	050_NZOJ_10F_11	16	R	H9
GCCTTTCA	IGVFSM4889EESW	050_NZOJ_10F_11	16	T	H9
TATGCTTC	IGVFSM7348XFYJ	062_WSBJ_10F_10	16	R	H9
GCCTTTCA	IGVFSM7348XFYJ	062_WSBJ_10F_10	16	T	H9
TATGCTTC	IGVFSM0359CZGZ	062_WSBJ_10F_11	16	R	H9
GCCTTTCA	IGVFSM0359CZGZ	062_WSBJ_10F_11	16	T	H9

ATGGTGTT	IGVFSM2896IHAT	051_NZOJ_10M_10	16	R	H10
ATTCTAGG	IGVFSM2896IHAT	051_NZOJ_10M_10	16	T	H10
ATGGTGTT	IGVFSM0427HINK	051_NZOJ_10M_11	16	R	H10
ATTCTAGG	IGVFSM0427HINK	051_NZOJ_10M_11	16	T	H10
ATGGTGTT	IGVFSM4977BSZT	061_WSBJ_10M_10	16	R	H10
ATTCTAGG	IGVFSM4977BSZT	061_WSBJ_10M_10	16	T	H10
ATGGTGTT	IGVFSM1160IHPD	061_WSBJ_10M_11	16	R	H10
ATTCTAGG	IGVFSM1160IHPD	061_WSBJ_10M_11	16	T	H10
GAATAATG	IGVFSM8339CJUO	052_NZOJ_10F_10	16	R	H11
CCTTACAT	IGVFSM8339CJUO	052_NZOJ_10F_10	16	T	H11
GAATAATG	IGVFSM3012TVCO	052_NZOJ_10F_11	16	R	H11
CCTTACAT	IGVFSM3012TVCO	052_NZOJ_10F_11	16	T	H11
GAATAATG	IGVFSM2781SEWP	096_WSBJ_10F_10	16	R	H11
CCTTACAT	IGVFSM2781SEWP	096_WSBJ_10F_10	16	T	H11
GAATAATG	IGVFSM0153ZUFQ	096_WSBJ_10F_11	16	R	H11
CCTTACAT	IGVFSM0153ZUFQ	096_WSBJ_10F_11	16	T	H11
CCTCTGTG	IGVFSM2662INEM	053_NZOJ_10M_10	16	R	H12
ACATTTGG	IGVFSM2662INEM	053_NZOJ_10M_10	16	T	H12
CCTCTGTG	IGVFSM7621BJTR	053_NZOJ_10M_11	16	R	H12
ACATTTGG	IGVFSM7621BJTR	053_NZOJ_10M_11	16	T	H12
CCTCTGTG	IGVFSM4132OPDD	063_WSBJ_10M_10	16	R	H12
ACATTTGG	IGVFSM4132OPDD	063_WSBJ_10M_10	16	T	H12
CCTCTGTG	IGVFSM9229AVFG	063_WSBJ_10M_11	16	R	H12
ACATTTGG	IGVFSM9229AVFG	063_WSBJ_10M_11	16	T	H12

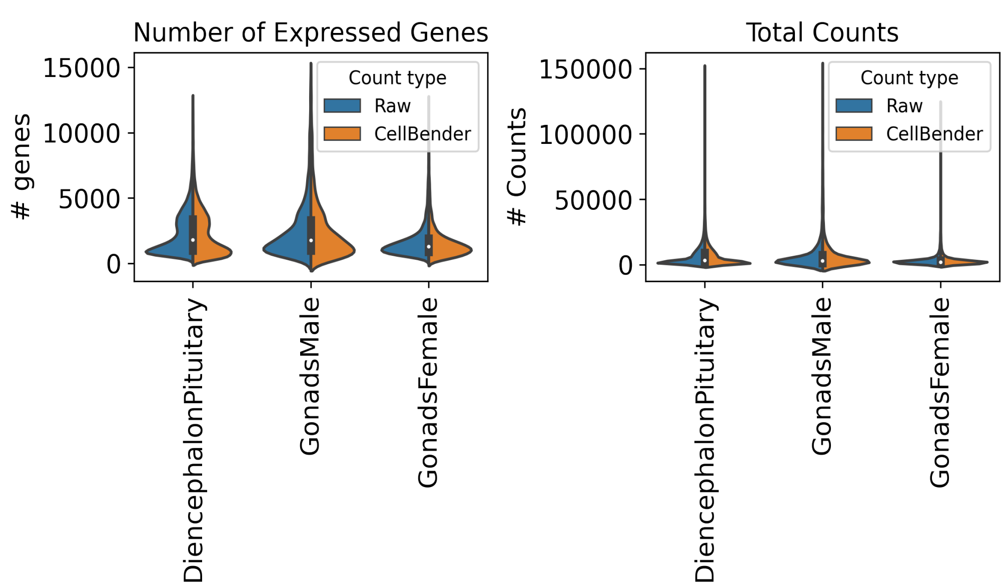
6. Description of data processing workflow and h5ad files

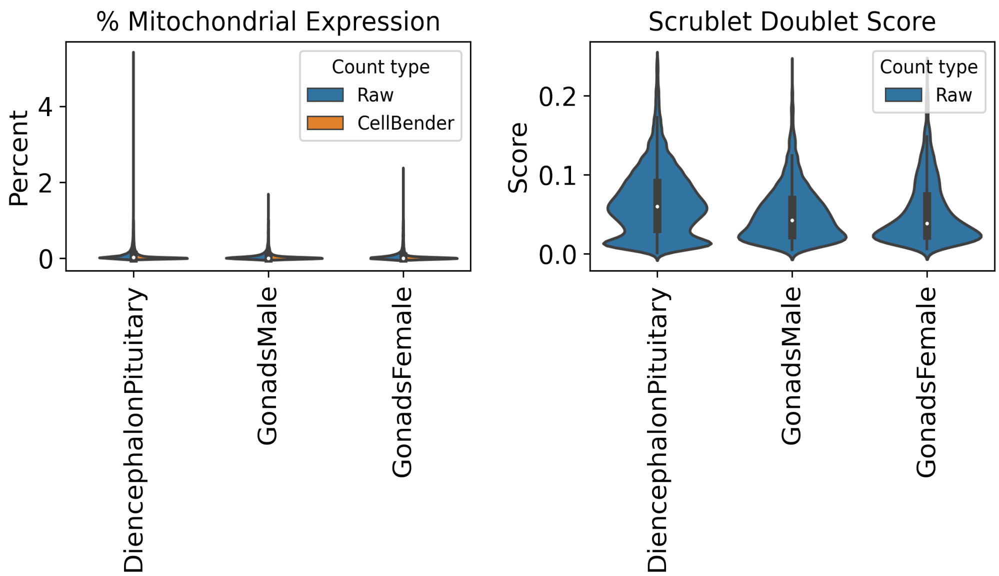
Data from Subpool_14 were combined with all other subpools into one AnnData object per tissue as well as matching data from other plates for processing and cell type annotation. The AnnData contains detailed sample and cell metadata in the observation (obs) table and gene information in the variables table (var). Cell IDs (adata.obs.index) are re-formatted to be human-readable and unique across subpools and experiments by appending subpool and experiment IDs. The final AnnData contains both raw and CellBender denoised counts matrices in separate layers: adata.layers['raw_counts'] and adata.layers['cellbender_counts']. The current adata.X points to the CellBender denoised counts. Cells or nuclei are filtered by >0.5 cell_probability from CellBender analysis.

In addition to CellBender filtering, nuclei were also filtered by the following QC parameters:

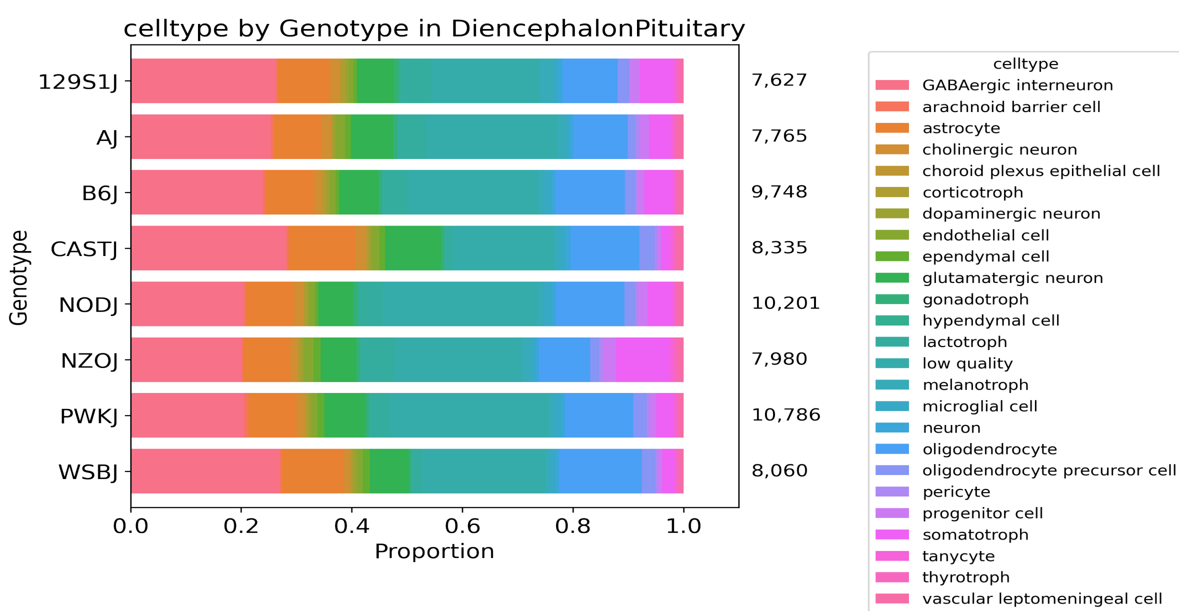
Table 8: QC thresholds

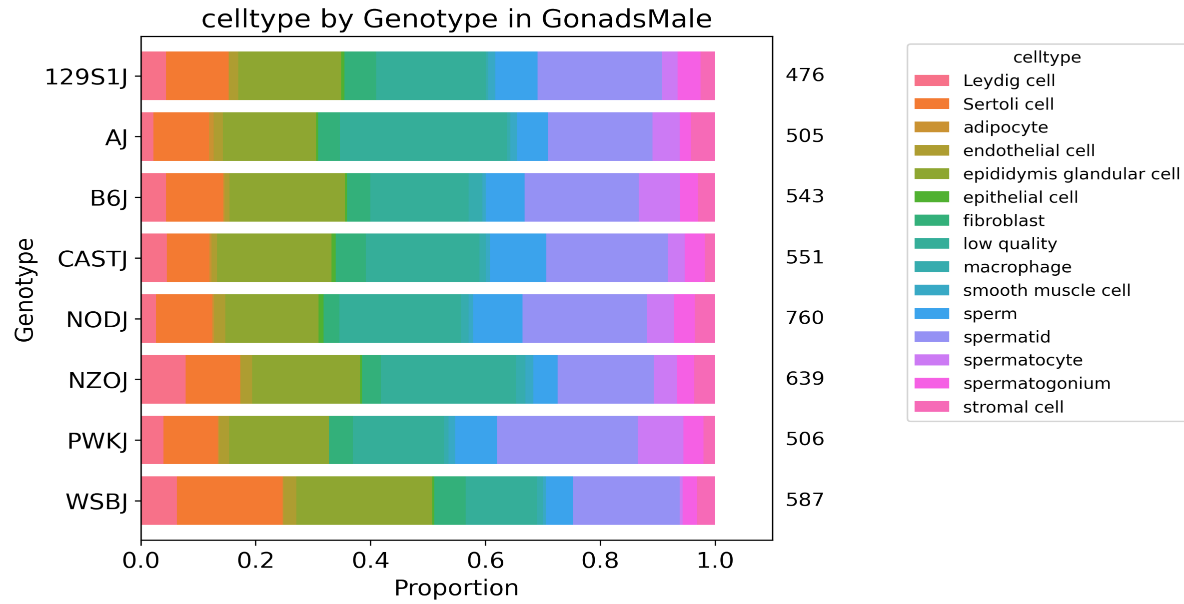
Parameter	Threshold
Minimum # UMIs	500.0
Maximum # UMIs	150000.0
Minimum # genes	250.0
Maximum % mito.	1.0
Maximum doublet score	0.25





After filtering, data for each tissue was normalized and clustered using CellBender denoised counts. Normalization includes regression of technical parameters (number of genes expressed per cell and percent mitochondrial gene expression). Harmony was used to integrate exome capture and non-exome capture subpools for annotation. A Leiden clustering resolution of 1 was used for 55 total clusters in DiencephalonPituitary, 45 clusters in GonadsMale, and 40 clusters in GonadsFemale using all tissue data across experiments. Cell type annotations were assigned per Leiden cluster or subcluster (leiden_R) using expression of canonical cell type marker genes.





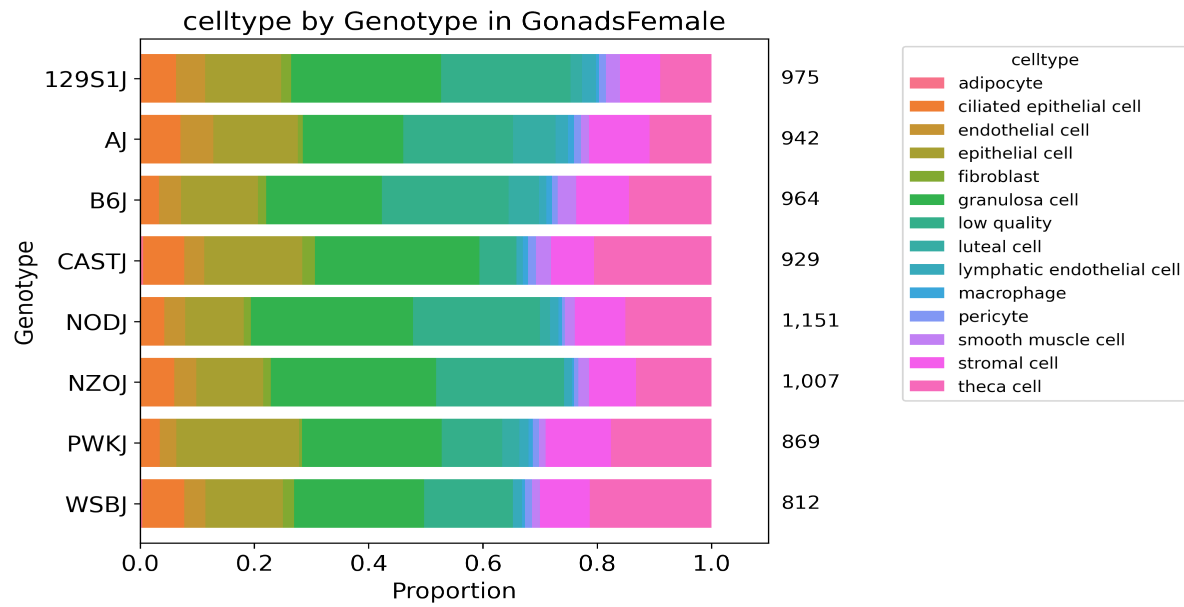


Table 9: Description of obs keys in h5ad file

Key	Description
cellID	Index of obs table. Includes barcoding wells for each round (1_2_3), subpool, and experiment.
lab_sample_id	Human-readable sample name used in-house.
sample	IGVF tissue accession.
plate	Experiment ID, e.g. igvf_003.
subpool	Subpool ID, e.g. Subpool_2. Number corresponds to Illumina index ID from Parse Bio kit.

SampleType	Nuclei or Cells.
Tissue	Human-readable tissue name, e.g. Kidney.
Sex	Male or Female.
Age	Age in postnatal months (PNM) of the mouse, e.g. PNM_02.
Genotype	Mouse genotype/strain, e.g. B6J.
leiden	Numeric Leiden cluster starting from 0.
leiden_R	If necessary, Leiden clusters and any sub-clusters, e.g. 15_1.
subpool_type	Exome capture (EX) or no exome capture (NO).
general_celltype	Broadest level of cell type annotations, e.g. neuron.
general_CL_ID	EMBL CL database ID for general cell type annotation, e.g. CL:0000540.
celltype	Finest level of cell type annotation with a CL ID, e.g. Vip+ GABAergic cortical interneuron.
CL_ID	EMBL CL database ID for cell type annotation, CL:4023016
subtype	Finest level of cell type annotation that may not have a CL ID. Mostly matches celltype.
Protocol	Cell barcoding and library building protocol/kit, e.g. Parse_WT.
Chemistry	Protocol chemistry version, v2 or v3.
bc	24-nucleotide sequence corresponding to 3 8-nucleotide barcode rounds (3, 2, 1).
bc1_sequence	8-nucleotide sequence barcode for round 1.
bc2_sequence	8-nucleotide sequence barcode for round 2.
bc3_sequence	8-nucleotide sequence barcode for round 3.
bc1_well	Round 1 well.
bc2_well	Round 2 well.
bc3_well	Round 2 well.
Row	Sample barcoding plate (round 1) row.
Column	Sample barcoding plate (round 1) column.
well_type	Single or Multiplexed.
Mouse_Tissue_ID	Same as lab_sample_id.
Multiplexed_sample1	Sample 1 from multiplexed well (if applicable)
Multiplexed_sample2	Sample 2 from multiplexed well (if applicable)
DOB	Date of birth of the mouse in month/day/year format.
Age_days	Age of the mouse in days.
Body_weight_g	Body weight in grams of the mouse.
Estrus_cycle	Estimated estrus stage of the mouse, if applicable.
Dissection_date	Date the tissue was harvested in month/day/year format.
Dissection_time	Time in hours:minutes (24-hour time) the mouse was sacrificed.
ZT	Number of hours into the light cycle, from lights on (06:30, ZT0) to lights off (20:30, ZT14).

Dissector	Initials of technician who dissected the tissue.
Tissue_weight_mg	Tissue weight in milligrams.
Notes	Notes taken during sample collection, experiment, or data processing.
n_genes_by_counts_raw	The number of genes with at least 1 count, calculated using raw counts.
total_counts_raw	Total counts (UMIs), calculated using raw counts.
total_counts_mt_raw	Total counts for mitochondrial genes, calculated using raw counts.
pct_counts_mt_raw	Percent of total counts in mitochondrial genes, calculated using raw counts.
n_genes_by_counts_cb	The number of genes with at least 1 count, calculated using CellBender counts.
total_counts_cb	Total counts (UMIs), calculated using CellBender counts.
total_counts_mt_cb	Total counts for mitochondrial genes, calculated using CellBender counts.
pct_counts_mt_cb	Percent of total counts in mitochondrial genes, calculated using CellBender counts.
doublet_score	Scrublet doublet score, calculated using CellBender counts.
predicted_doublet	Scrublet predicted doublet (True or False).
background_fraction	Fraction of counts CellBender predicted to be background noise.
cell_probability	CellBender probability that the cell is real.
cell_size	CellBender size factor.
droplet_efficiency	CellBender statistic indicating transcript capture efficiency per cell.
B6J_klue_counts	Estimated counts for B6J genotype from klue analysis.
NODJ_klue_counts	Estimated counts for NODJ genotype from klue analysis.
AJ_klue_counts	Estimated counts for AJ genotype from klue analysis.
PWKJ_klue_counts	Estimated counts for PWKJ genotype from klue analysis.
129S1J_klue_counts	Estimated counts for 129S1J genotype from klue analysis.
CASTJ_klue_counts	Estimated counts for CASTJ genotype from klue analysis.
WSBJ_klue_counts	Estimated counts for WSBJ genotype from klue analysis.
NZOJ_klue_counts	Estimated counts for NZOJ genotype from klue analysis.

7. Package versions

Table 10: Package versions

package	version
kallisto	0.50.1
bustools	0.43.2
kb	0.28.2
Python	3.9.0
snakemake	7.32.0

cellbender	0.3.2
scanpy	1.10.2
scrublet	0.2.3
anndata	0.10.8
pandas	2.2.2
numpy	1.26.4
harmonypy	0.0.9

8. Contact information and GitHub

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Pipeline repo: https://github.com/mortazavilab/parse_pipeline

Code used to make this report: https://github.com/mortazavilab/8cube_manuscript/blob/main/plate_report/Subpool_report_igvf_007.ipynb

Code used to process and annotate the data for main tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/DiencephalonPituitary.ipynb

Code used to process and annotate the data for multiplexed tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/GonadsMale.ipynb